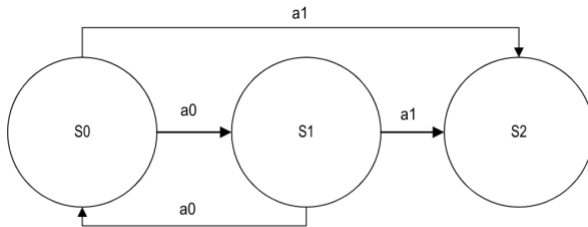


IT5005 Artificial Intelligence
Tutorial 9
SOLUTIONS

1. Consider the Markov Decision Process (MDP) shown below. All transitions are deterministic.

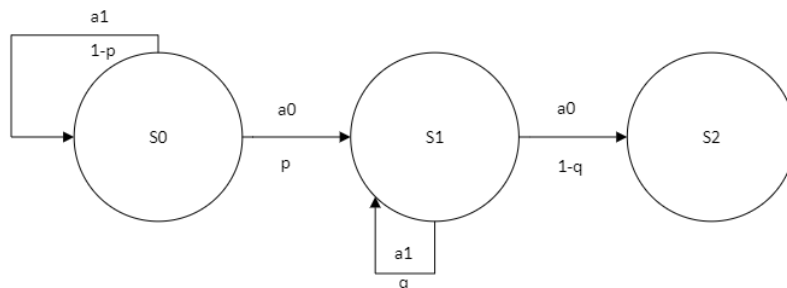


- List all the permitted actions in each state.
- List all the possible deterministic policies.
- Find the value of each state under each policy in part a. The rewards for each action and each state are shown below:

State	a0	a1
S_0	1	1
S_1	2	0
S_2	4	4

We assume that taking an action that is not valid for a particular state has no effect; i.e. we return to the same state. So taking any action in state S_2 will always return to state S_2 . Thus S_2 is a “sink” or “terminal” state, which is quite common since almost all MDPs would end in a terminal state. We assume that $0 \leq \gamma < 1$.

2. Consider now a probabilistic MDP as shown below. The actions are shown above the edges representing the actions and transitions, while the probabilities of taking those actions are shown below:



For example in state S_0 there is a probability p of taking action $a0$ and moving to state S_1 , and a probability $1 - p$ of taking action $a1$ and returning to state S_0 . As before S_2 is a terminal state and taking any action there will always return you back to S_2 .

Find the expression for the value of each state given the following rewards, and $0 \leq \gamma < 1$.

State	a0	a1
S_0	2	3
S_1	3	1
S_2	2	2

3. One urn contains 10 red balls and 25 green balls, and a second urn contains 22 red balls and 15 green balls. A ball is chosen as follows: First an urn is selected by tossing a loaded coin with probability 0.4 of landing heads up and probability of 0.6 of landing tails up. If the coin lands heads up, the first urn is chosen; otherwise, the second urn is chosen. Then a ball is picked at random from the chosen urn.

Write your answers correct to three significant figures.

- What is the probability that the chosen ball is green?
- If the chosen ball is green, what is the probability that it was picked from the first urn?